

### Given:

- **IP Address:** 10.0.5.17
- **Subnet Mask:** 255.255.255.0

### Step 1: Convert the Subnet Mask to CIDR Notation

- **255.255.255.0 = /24**

### Step 2: Find the Network Address

Using the **bitwise AND** of the IP address and subnet mask:

- **IP Address:** 10.0.5.17
- **Subnet Mask:** 255.255.255.0
- **Network Address:** **10.0.5.0**

### Step 3: Find the Broadcast Address

For a /24 subnet, the last address in the subnet is the broadcast address.

- **Broadcast Address:** **10.0.5.255**

### Step 4: Find the Valid Host Range

- **First Usable Host:** **10.0.5.1**
- **Last Usable Host:** **10.0.5.254**

| Item              | Value                 |
|-------------------|-----------------------|
| IP Address        | 10.0.5.17             |
| Subnet Mask       | 255.255.255.0         |
| CIDR Notation     | /24                   |
| Network Address   | 10.0.5.0              |
| Broadcast Address | 10.0.5.255            |
| Valid Host Range  | 10.0.5.1 – 10.0.5.254 |